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MULTIVARIATE:-

PAST PAPERS:-

— (2017) —

— LONG QUESTIONS —

— Q. NO. 3 —

Find the multivariate M.G.F
of X with mean $\underline{\mu}$ and
variance Σ .

Proof:-

As

$$M_X(t) = E[e^{t'X}]$$

$$\because X \sim N_p(\underline{\mu}, \Sigma)$$

And

$$f(x) = \frac{1}{\Sigma^{1/2} (2\pi)^{p/2}} \exp\left[-\frac{1}{2} \{(\underline{x} - \underline{\mu})' \Sigma^{-1} (\underline{x} - \underline{\mu})\} \right]$$

$-\infty < x_i < \infty$

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According to positive definite matrix Σ^{-1} , there exist a symmetric non-singular matrix C such that.

$$C' \Sigma^{-1} C = I$$

taking mod on b/s.

$$|C' \Sigma^{-1} C| = |I|$$

$$|C'| |\Sigma^{-1}| |C| = 1$$

$$|C|^2 |\Sigma^{-1}| = 1$$

$$|C|^2 = \frac{1}{|\Sigma^{-1}|}$$

$$|C|^2 = |\Sigma^{-1}|^{-1} = |\Sigma|$$

taking square root on b/s.

$$|C| = |\Sigma|^{1/2}$$

$$\boxed{\text{mod } |C| = |\Sigma|^{1/2}}$$

let

$(X - \mu) = CZ$ from the functions by Jacobian method.

$$X = CZ + \mu$$

$$J = \frac{\partial X}{\partial Z} = C + 0$$

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$$J = C$$

$$|J| = \text{mod } |C|$$

Now,

$$(\underline{x} - \underline{\mu})' \underline{\Sigma}^{-1} (\underline{x} - \underline{\mu}) = (\underline{C}\underline{Z})' \underline{\Sigma}^{-1} (\underline{C}\underline{Z})$$

$$= \underline{C}' \underline{Z}' \underline{\Sigma}^{-1} \underline{C} \underline{Z}$$

$$= \underline{Z}' \underline{C}' \underline{\Sigma}^{-1} \underline{C} \underline{Z}$$

$$= \underline{Z}' \underline{I} \underline{Z}$$

$$= \underline{Z}' \underline{Z}$$

$$= \sum_{i=1}^p \underline{Z}_i^2$$

Now the p.d.f of multivariate normal dist becomes convert

$\underline{x}$ into $\underline{z}$.

$$f(\underline{z}) = \frac{1}{|C|^{1/2} (2\pi)^{p/2}} e^{-\frac{1}{2} \sum_{i=1}^p \underline{Z}_i^2} |J|$$

$$= \frac{1}{|C|^{1/2} (2\pi)^{p/2}} e^{-\frac{1}{2} \sum_{i=1}^p \underline{Z}_i^2} \text{mod } |C|$$

$$= \frac{1}{|C|^{1/2} (2\pi)^{p/2}} e^{-\frac{1}{2} \sum_{i=1}^p \underline{Z}_i^2} |C|^{1/2}$$

$$= \frac{1}{|C|^{1/2} (2\pi)^{p/2}} e^{-\frac{1}{2} \sum_{i=1}^p \underline{Z}_i^2} |C|^{1/2}$$

$$= \frac{1}{|C|^{1/2} (2\pi)^{p/2}} e^{-\frac{1}{2} \sum_{i=1}^p \underline{Z}_i^2}$$

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Dr. P. S. Vaidyanathan P. S. V.

$$f(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2\sigma^2}}$$

Now the MGF of Z is

$$M_Z(t) = E(e^{tZ}) = \int_{-\infty}^{\infty} e^{tZ} f(Z) dZ$$

$$M_Z(t) = \frac{1}{\sigma\sqrt{2\pi}} E(e^{tZ}) \quad (2)$$

Now,

$$E(e^{tZ}) = \int_{-\infty}^{\infty} e^{tZ} f(Z) dZ$$

$$= \int_{-\infty}^{\infty} e^{tZ} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{Z^2}{2}} dZ \quad \text{where, } Z \sim N(0,1)$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{Z^2}{2} + tZ} dZ$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(Z^2 - 2tZ)} dZ$$

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$$\begin{aligned}
 E(e^{t_1 z_1}) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z_1^2 - 2t_1 z_1 + t_1^2 - t_1^2)} dz_1 \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}[(z_1^2 - 2t_1 z_1 + t_1^2) - t_1^2]} dz_1 \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z_1^2 - 2t_1 z_1 + t_1^2)} \cdot e^{t_1^2/2} dz_1 \\
 &= \frac{e^{t_1^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(z_1 - t_1)^2} dz_1
 \end{aligned}$$

Let

$$z_1 - t_1 = w_1$$

$$dz_1 = dw_1$$

limit $-\infty$ to ∞

$$E(e^{t_1 z_1}) = \frac{e^{t_1^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}w_1^2} dw_1$$

$$= e^{t_1^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{w_1^2}{2}} dw_1$$

% Area under the curve of Normal is Unity.

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{w_1^2}{2}} dw_1 = 1$$

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So,

$$E(e^{t_1 z_1}) = e^{t_1^2/2} (1)$$

$$E(e^{t_1 z_1}) = e^{t_1^2/2}$$

put in equation (2)

$$M_z(t) = \prod_{i=1}^p E(e^{t_i z_i})$$

$$= e^{\sum_{i=1}^p t_i^2/2}$$

$$\because \underline{t}^T \underline{t} = t_1^2 + t_2^2 + t_3^2 + \dots + t_p^2 = \sum_{i=1}^p t_i^2$$

$$M_z(t) = e^{\underline{t}^T \underline{t}/2}$$

Now From eq (1)

$$M_x(t) = E(e^{\underline{t}^T \underline{x}})$$

$$= E[e^{\underline{t}^T (\underline{u} + \underline{C} \underline{z})}]$$

$$\because \underline{x} = \underline{u} + \underline{C} \underline{z}$$

$$= E[e^{\underline{t}^T \underline{u} + \underline{t}^T \underline{C} \underline{z}}]$$

$$= e^{\underline{t}^T \underline{u}} E(e^{\underline{t}^T \underline{C} \underline{z}})$$

$$\because M_z(t) = e^{\underline{t}^T \underline{t}/2}$$

$$M_z(\underline{C}^T \underline{t}) = e^{\underline{C}^T \underline{t}^T \underline{C} \underline{t}/2}$$

So,

$$= e^{\underline{t}^T \underline{u}} M_z(\underline{C}^T \underline{t})$$

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$$\underline{W} = \sum_{i=1}^n \underline{x}_i \underline{x}_i'$$

where,

$$\underline{x}_i \sim N(\underline{\mu}_i, \underline{\Sigma})$$

taking

$$\underline{C}' \underline{W} \underline{C} = \sum_{i=1}^n \underline{C}' \underline{x}_i \underline{x}_i' \underline{C}$$

$$\underline{C}' \underline{W} \underline{C} = \sum_{i=1}^n (\underline{C}' \underline{x}_i) (\underline{C}' \underline{x}_i)'$$

let

$$\underline{C}' \underline{x}_i = \underline{z}_i$$

so,

$$\underline{C}' \underline{W} \underline{C} = \sum_{i=1}^n \underline{z}_i \underline{z}_i'$$

where

$$\underline{z}_i \sim N(\underline{C}' \underline{\mu}_i, \underline{C}' \underline{\Sigma} \underline{C})$$

$$\begin{aligned} \therefore E(\underline{z}_i) &= E(\underline{C}' \underline{x}_i) \\ &= \underline{C}' E(\underline{x}_i) \\ &= \underline{C}' \underline{\mu}_i \end{aligned}$$

And

$$\begin{aligned} \therefore \text{cov}(\underline{z}_i) &= \text{cov}(\underline{C}' \underline{x}_i) \\ &= \underline{C}' \text{cov}(\underline{x}_i) \underline{C} \\ &= \underline{C}' \underline{\Sigma} \underline{C} \end{aligned}$$

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$$M_X(t) = e^{-\frac{t'u}{2}} e^{\frac{1}{2}(\underline{C}'t)'(\underline{C}t)}$$

$$= e^{-\frac{t'u}{2}} e^{\frac{1}{2}t' \underline{C}' \underline{C} t}$$

$$= e^{-\frac{t'u}{2}} e^{\frac{1}{2}t' \underline{\Sigma} t}$$

$$\because \underline{C}' \underline{\Sigma}^{-1} \underline{C} = \underline{I}$$

$$\underline{C}' \underline{C} = \underline{\Sigma}$$

$$M_X(t) = e^{-\frac{t'u}{2} + \frac{1}{2}t' \underline{\Sigma} t}$$

It is the M.G.F of 'X'
In multivariate Normal Dist.

Q. NO. 4

Prove the linear combination of
Wishart. $W \sim W_p(\underline{F}; \underline{\Sigma}, \underline{M})$ then show
 $\underline{C}'W\underline{C} \sim W_q(\underline{F}, \underline{C}'\underline{\Sigma}\underline{C}, \underline{M}\underline{C})$

Proof:-

Since,

$$W \sim W_p(\underline{F}, \underline{\Sigma}, \underline{M})$$

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Hence

$$\underline{C}' \underline{W} \underline{C} = \sum_{i=1}^p \underline{z}_i \underline{z}_i' \sim W_p(p, \underline{C}' \underline{C}, N)$$

where Non-Centrality parameter = $\underline{N}$

$$\underline{N} = \begin{pmatrix} (\underline{C}' \underline{u}_1)' \\ (\underline{C}' \underline{u}_2)' \\ \vdots \\ (\underline{C}' \underline{u}_p)' \end{pmatrix} = \begin{pmatrix} \underline{C}' \underline{u}_1 \\ \underline{C}' \underline{u}_2 \\ \vdots \\ \underline{C}' \underline{u}_p \end{pmatrix}$$

$$= \underline{C} \begin{pmatrix} \underline{u}_1' \\ \underline{u}_2' \\ \vdots \\ \underline{u}_p' \end{pmatrix}$$

$$\underline{N} = \underline{C} \underline{M}$$

Now

$$\underline{C}' \underline{W} \underline{C} \sim W_p(p, \underline{C}' \underline{C}, \underline{M} \underline{C})$$

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— of Q. NO. 5) —

Show that

$3x_1^2 + 2x_2^2 - 2\sqrt{2}x_1x_2$
is positive definite.

Solution,

As, let $x = \begin{bmatrix} x_1 & x_2 \end{bmatrix}$

The combination of x_1 and x_2 is

$$\begin{bmatrix} x_1x_1 & x_1x_2 \\ x_2x_1 & x_2x_2 \end{bmatrix} = \begin{bmatrix} x_1^2 & x_1x_2 \\ x_2x_1 & x_2^2 \end{bmatrix}$$

$$\Rightarrow x_2x_1 = x_1x_2$$

$$Q(x) = x'Ax$$

$$\text{So, } A = \begin{bmatrix} 3 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{bmatrix}$$

The characteristics eq of A is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 3-\lambda & -\sqrt{2} \\ -\sqrt{2} & 2-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)(2-\lambda) - 2 = 0$$

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$$\lambda^2 - 5\lambda + 4 = 0$$

$$\lambda - 4\lambda - \lambda + 4 = 0$$

$$\lambda(\lambda - 4) - 1(\lambda - 4) = 0$$

$$(\lambda - 4)(\lambda - 1) = 0$$

$$\boxed{\lambda_1 = 4}$$

$$\boxed{\lambda_2 = 1}$$

Now, the spectral decomposition
of matrix A

$$A = \lambda_1 \underline{e}_1 \underline{e}_1' + \lambda_2 \underline{e}_2 \underline{e}_2'$$

$$= 4 \underline{e}_1 \underline{e}_1' + 1 \underline{e}_2 \underline{e}_2'$$

$$A = 4 \underline{e}_1 \underline{e}_1' + \underline{e}_2 \underline{e}_2'$$

Pre multiplying by $\underline{x}'$ and post
multiplying by $\underline{x}$

$$\underline{x}' A \underline{x} = 4 \underline{x}' \underline{e}_1 \underline{e}_1' \underline{x} + \underline{x}' \underline{e}_2 \underline{e}_2' \underline{x}$$

let

$$y_1 = \underline{x}' \underline{e}_1 = \underline{e}_1' \underline{x}$$

$$y_2 = \underline{x}' \underline{e}_2 = \underline{e}_2' \underline{x}$$

So,

$$\underline{x}' A \underline{x} = 4 y_1 y_1 + y_2 y_2$$

$$= 4 y_1^2 + y_2^2 \geq 0 \rightarrow \text{A}$$

In matrix form.

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$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} e'_1 x \\ e'_2 x \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} e'_1 \\ e'_2 \end{bmatrix} x$$

$$\underline{y} = \underline{E} \underline{x}$$

$$\underline{x} = \underline{E}' \underline{y}$$

As $\underline{E}$ is a orthogonal matrix
hence has a inverse as $\underline{E}'$.

But

$$\underline{x} \neq 0, \underline{y} \neq 0$$

So from eq (A)

$$\underline{x}' \underline{A} \underline{x} = 4y_1^2 + y_2^2 > 0$$

i.e the given quadratic
form and matrix $\underline{A}$
is positive definite.

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Q. NO. 6) B2

Find

a) $E[Y_1/Y_2]$

b) $\text{COV}[Y_1/Y_2]$

Given

the random vector $U = \begin{bmatrix} 4 \\ 0 \\ -3 \\ 7 \end{bmatrix}$

$X = [X_1 \ X_2 \ X_3 \ X_4]$
 $\Sigma = \begin{bmatrix} 7 & -1 & 0 & 3 \\ -1 & 8 & -2 & 6 \\ 0 & -2 & 12 & 9 \\ 3 & 6 & 9 & 3 \end{bmatrix}$

Suppose

$Y_1' = [X_1 \ X_2]$

$Y_2' = [X_3 \ X_4]$

Solution:

$U = \begin{bmatrix} 4 \\ 0 \\ -3 \\ 7 \end{bmatrix} \Rightarrow \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{matrix}$, $\Sigma = \begin{bmatrix} 7 & -1 & 0 & 3 \\ -1 & 8 & -2 & 6 \\ 0 & -2 & 12 & 9 \\ 3 & 6 & 9 & 3 \end{bmatrix}$

Now,

$\Sigma_{11} = \begin{bmatrix} 7 & -1 \\ -1 & 8 \end{bmatrix}$, $\Sigma_{12} = \begin{bmatrix} 0 & 3 \\ -2 & 6 \end{bmatrix}$

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$$\underline{\Sigma}_{21} = \begin{bmatrix} 0 & -8 \\ 3 & 6 \end{bmatrix}, \quad \underline{\Sigma}_{22} = \begin{bmatrix} 12 & 9 \\ 9 & 3 \end{bmatrix}$$

Find $\underline{\Sigma}_{22}^{-1} = ?$

$$|\underline{\Sigma}_{22}| = \begin{vmatrix} 12 & 9 \\ 9 & 3 \end{vmatrix} = 36 - 81$$

$$|\underline{\Sigma}_{22}| = -45$$

$$\text{Adj } \underline{\Sigma}_{22} = \begin{bmatrix} 3 & -9 \\ -9 & 12 \end{bmatrix}$$

$$\underline{\Sigma}_{22}^{-1} = \frac{\text{Adj } \underline{\Sigma}_{22}}{|\underline{\Sigma}_{22}|}$$
$$= \frac{\begin{bmatrix} 3 & -9 \\ -9 & 12 \end{bmatrix}}{-45}$$

$$= \begin{bmatrix} -3/45 & 9/45 \\ 9/45 & -12/45 \end{bmatrix}$$

$$\underline{\Sigma}_{22}^{-1} = \begin{bmatrix} -0.06 & 0.2 \\ 0.2 & -0.26 \end{bmatrix}$$

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a) $E(Y_1/Y_2) = ?$

As $E(Y_1/Y_2) = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (Y_2 - \mu_2)$

$$= \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 3 \\ -2 & 6 \end{bmatrix} \begin{bmatrix} -0.06 & 0.2 \\ 0.2 & -0.26 \end{bmatrix} \left[Y_2 - \begin{bmatrix} -3 \\ 7 \end{bmatrix} \right]$$

$$= \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.6 & -0.78 \\ 1.32 & -1.96 \end{bmatrix} \begin{bmatrix} Y_2 + 3 \\ Y_2 - 7 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} -0.18Y_2 + 6.48 \\ -0.64Y_2 + 17.68 \end{bmatrix}$$

$$E(Y_1/Y_2) = \begin{bmatrix} -0.18Y_2 + 10.48 \\ -0.64Y_2 + 17.68 \end{bmatrix}$$

b) $\text{COV}(Y_1/Y_2) = ?$

$$\text{COV}(Y_1/Y_2) = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

$$= \begin{bmatrix} 7 & -1 \\ -1 & 8 \end{bmatrix} - \begin{bmatrix} 0 & 3 \\ -2 & 6 \end{bmatrix} \begin{bmatrix} -0.06 & 0.2 \\ 0.2 & -0.26 \end{bmatrix} \begin{bmatrix} 0 & -2 \\ 3 & 6 \end{bmatrix}$$

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$$= \begin{bmatrix} 7 & -1 \\ -1 & 8 \end{bmatrix} - \begin{bmatrix} 0.6 & -0.78 \\ 1.32 & -1.96 \end{bmatrix} \begin{bmatrix} 0 & -2 \\ 3 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & -1 \\ -1 & 8 \end{bmatrix} - \begin{bmatrix} -2.34 & -5.88 \\ -5.88 & -14.4 \end{bmatrix}$$

$$\text{Cov}(y_1, y_2) = \begin{bmatrix} 9.34 & 4.88 \\ 4.88 & 22.4 \end{bmatrix}$$



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LONG QUESTIONS

(Q.NO.3)

Find 'K' of following density function.

$$f(x) = K \exp\left[-\frac{1}{2}\{(x-\mu)' \Sigma^{-1}(x-\mu)\}\right]$$

And Also $E(X) = ?$

Solution:

Since the density function is given that $f(x)$ is

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x) dx_1 \dots dx_p = 1$$

$$\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x) dx_1 \dots dx_p = 1$$

$$\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} K \exp\left[-\frac{1}{2}\{(x-\mu)' \Sigma^{-1}(x-\mu)\}\right] dx_1 \dots dx_p = 1$$

← (1)

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According to positive definite matrix Σ^{-1} there exist a symmetric non-singular matrix C such that

$$\underline{C}' \underline{\Sigma}^{-1} \underline{C} = \underline{I}$$

taking mod on b/s

$$|\underline{C}' \underline{\Sigma}^{-1} \underline{C}| = 1$$

$$|\underline{C}|^2 = |\underline{\Sigma}|$$

$$\therefore \text{mod } |\underline{C}| = |\underline{\Sigma}|^{1/2}$$

And let

$$\underline{X} - \underline{u} = \underline{C} \underline{Z}$$

$$(\underline{X} - \underline{u})' \underline{\Sigma}^{-1} (\underline{X} - \underline{u}) = (\underline{C} \underline{Z})' \underline{\Sigma}^{-1} (\underline{C} \underline{Z})$$

$$= \underline{Z}' \underline{C}' \underline{\Sigma}^{-1} \underline{C} \underline{Z}$$

$$= \underline{Z}' \underline{I} \underline{Z}$$

$$= \underline{Z}' \underline{Z}$$

$$= \sum_{i=1}^p z_i^2$$

$$\underline{X} = \underline{C} \underline{Z} + \underline{u}$$

Jacobiel method

$$\underline{J} = \frac{\partial \underline{X}}{\partial \underline{Z}} = \underline{C}$$

$$\underline{J}$$

$$|\underline{J}| = \text{mod } |\underline{C}|$$

Now From equation (1).

$$K \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{2} \sum_{i=1}^p z_i^2} |\underline{J}| dz_i$$

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$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \text{mod } |C| = \frac{1}{|C|} \quad (2)$$

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} |C|^{1/2} = \frac{1}{|C|}$$

$$\therefore \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} = 1$$

$$\int_{-\infty}^{\infty} e^{-z^2/2} = \sqrt{2\pi}$$

So, eq (2) becomes.

$$\frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} |C|^{1/2} = \frac{1}{|C|}$$

$$(2\pi)^{1/2} |C|^{1/2} = \frac{1}{|C|}$$

$$|C| = \frac{1}{|C|^{1/2} (2\pi)^{1/2}}$$

$$P(x) = ?$$

As $X = CZ + \mu$

$$E(X) = \mu + C E(Z)$$

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Now the p.d.f of multivariate normal dist is:

$$f(\underline{x}) = \frac{1}{|\underline{\Sigma}|^{1/2} (2\pi)^{p/2}} \exp\left[-\frac{1}{2}(\underline{x}-\underline{\mu})' \underline{\Sigma}^{-1}(\underline{x}-\underline{\mu})\right]$$

$$= \frac{1}{|\underline{\Sigma}|^{1/2} (2\pi)^{p/2}} \exp\left[-\frac{1}{2} \sum_{i=1}^p z_i^2\right] \cdot |\underline{\Sigma}|^{1/2}$$

By given Jacobian method and previous we get.

$$f(\underline{z}) = (2\pi)^{p/2} e^{-\frac{1}{2} \sum_{i=1}^p z_i^2}$$
$$= (2\pi)^{p/2} \cdot \frac{1}{\sqrt{\pi}} e^{-z_i^2/2}$$

So,

$$E(\underline{z}) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \underline{z} f(\underline{z}) dz_1 \dots dz_p$$

$$= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \underline{z} \cdot \frac{1}{\sqrt{\pi}} (2\pi)^{-p/2} e^{-\underline{z}^2/2} dz_1 \dots dz_p$$

$$= \int_{-\infty}^{\infty} z_i (2\pi)^{-p/2} e^{-z_i^2/2} dz_i \cdot \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} (2\pi)^{-p/2} e^{-z_j^2/2} dz_j \dots dz_p$$

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$$F(z) = (0) (1)^p$$

$$= 0$$

Generalize it.

$$F(z) = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_p \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \underline{0}$$

Put it in eq

$$F(x) = M + c F(z)$$

$$= U + 0$$

$$\boxed{F(x) = \underline{U}}$$

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Q. NO. 4

Prove the Linear Compound of
Wishart distⁿ $W \sim W_p(\underline{\Gamma}, \underline{\Sigma}, M)$
then show that
 $C'WC \sim \partial^2 \chi^2(\underline{\Gamma}, \underline{\delta}^2)$

Proof:

where,

$$C'WC \sim \partial^2 \chi^2(\underline{\Gamma}, \underline{\delta}^2) \text{ (non central)} \\ \text{wishart}$$

$$\partial^2 = C'WC$$

and

$$\partial^2 \underline{\delta}^2 = C' M M C$$

$\underline{\delta}$ is non-central χ^2 -parameter.

Proof:

$$\text{If } W \sim W_p(\underline{\Gamma}, \underline{\Sigma}, M)$$

then

$$W = \sum_{i=1}^M x_i x_i'$$

$$x_i \sim N_p(\mu_i, \underline{\Sigma})$$

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Consider,

$$\underline{C}' \underline{\omega} \underline{C} = \sum_{i=1}^p \underline{C}' x_i x_i' \underline{C}$$

$$= \sum_{i=1}^p (\underline{C}' x_i) (\underline{C}' x_i)'$$

As $\underline{C}' x_i$ is a scalar quantity.

$$\underline{C}' \underline{\omega} \underline{C} = \sum_{i=1}^p (\underline{C}' x_i) (\underline{C}' x_i)$$

$$\underline{C}' \underline{\omega} \underline{C} = \sum_{i=1}^p (\underline{C}' x_i)^2$$

(A)

Now,

$$x_i \sim N_p(\mu, \Sigma)$$

$$\underline{C}' x_i \sim N_q(\underline{C}' \mu, \underline{C}' \Sigma \underline{C})$$

As,

$$\frac{SS}{Var} \sim \chi^2(p, \delta^2)$$

$$\frac{SS}{\underline{C}' \underline{\omega} \underline{C}} \sim \chi^2(p, \delta^2)$$

$$\frac{\underline{C}' \underline{\omega} \underline{C}}{\delta^2} \sim \chi^2(p, \delta^2)$$

$$\underline{C}' \underline{\omega} \underline{C} \sim \delta^2 \chi^2(p, \delta^2)$$

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Now, Consider

$$\delta^2 = \sum_{i=1}^f (c' u_i)^2$$

$$\delta^2 = \sum_{i=1}^f (c' u_i) (c' u_i)$$

As it is a scalar quantity

$$\delta^2 = \sum_{i=1}^f (c' u_i) (c' u_i)'$$

$$= \sum_{i=1}^f c' u_i u_i' c$$

$$\delta^2 = c' \left(\sum_{i=1}^f u_i u_i' \right) c \quad \text{--- (B)}$$

As,

$$\underline{M} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_f \end{bmatrix}$$

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$$\underline{M}' = [u_1 \ u_2 \ u_3 \ \dots \ u_f]$$

$$\underline{M}' \underline{M} = [u_1 \ u_2 \ \dots \ u_f] \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_f \end{bmatrix}$$

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$$M'M = [u_1 u_1' + u_2 u_2' + \dots + u_f u_f']$$

$$M'M = \sum_{i=1}^f u_i u_i'$$

Equation 'B' becomes.

$$\delta^2 = C'MM'C$$

Hence proved,

$$C'WC \sim \delta^2 \chi^2 (f, \delta^2)$$

And

$$\delta^2 = C'MM'C$$

It is a linear compound.

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Q. NO. 5

Find the Spectral decomposition of matrix.

$$A = \begin{pmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{pmatrix}$$

Sol.

the characteristic eq. is

$$|A - \lambda I| = 0$$

$$\left| \begin{pmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right| = 0$$

$$\left| \begin{pmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = 0$$

$$\begin{vmatrix} 13-\lambda & -4 & 2 \\ -4 & 13-\lambda & -2 \\ 2 & -2 & 10-\lambda \end{vmatrix} = 0$$

ate:

$$(13-\lambda) \begin{vmatrix} 13-\lambda & -2 & +4 \\ -2 & 10-\lambda & 2 \end{vmatrix} + 4 \begin{vmatrix} -4 & -2 \\ 2 & 10-\lambda \end{vmatrix} + 2 \begin{vmatrix} +2 & -16 \\ 2 & -2 \end{vmatrix} = 0$$

$$13-\lambda \left[(13-\lambda)(10-\lambda) - 4 \right] + 4 \left[-40 + 4\lambda + 4 \right] + 2 \left[8 - 26 + 2\lambda \right] = 0$$

$$(13-\lambda) \left[130 - 13\lambda - 10\lambda + \lambda^2 - 4 \right] + \left[-160 + 16\lambda + 16 \right] + \left[16 - 20 + 4\lambda \right] = 0$$

$$(13-\lambda)(\lambda^2 - 23\lambda + 126) + (16\lambda - 144) + (4\lambda - 36) = 0$$

$$13\lambda^2 - 299\lambda + 1638 - \lambda^3 + 23\lambda^2 - 126\lambda + 16\lambda - 144 + 4\lambda - 36 = 0$$

$$-\lambda^3 + 36\lambda^2 - 405\lambda + 1458 = 0$$

$$\lambda^3 - 36\lambda^2 + 405\lambda - 1458 = 0$$

As the $\lambda = 9$ satisfied the above equation.

By using synthetic division.

$$\begin{array}{r|rrrr} & 1 & -36 & 405 & -1458 \\ 9 & 0 & 9 & -243 & 1458 \\ \hline & 1 & -27 & 162 & 0 \end{array}$$

$$(\lambda - 9)(\lambda^2 - 27\lambda + 162) = 0$$

$$\lambda - 9 = 0$$

$$\boxed{\lambda = 9}$$

$$\lambda^2 - 27\lambda + 162 = 0$$

$$\lambda^2 - 18\lambda - 9\lambda + 162 = 0$$

$$\lambda(\lambda - 18) - 9(\lambda - 18) = 0$$

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$$(\lambda - 18)(\lambda - 9) = 0$$

$$\lambda = 18, \lambda = 9$$

$$\lambda_1 = 18$$

$$\lambda_2, \lambda_3 = 9$$

Eigen Vector:-

For $\lambda_1 = 18$

$$A \underline{x} = \lambda_1 \underline{x}$$

$$\text{Let } \underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\begin{pmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 18 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$13x_1 - 4x_2 + 2x_3 = 18x_1$$

$$13x_1 - 4x_2 + 2x_3 - 18x_1 = 0$$

$$2x_3 - 4x_2 - 5x_1 = 0 \quad \text{--- (1)}$$

$$-4x_1 + 13x_2 - 2x_3 = 18x_2$$

$$-4x_1 + 13x_2 - 2x_3 - 18x_2 = 0$$

$$-2x_3 - 5x_2 - 4x_1 = 0 \quad \text{--- (2)}$$

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$$2x_1 - 2x_2 + 10x_3 - 18x_3 = 0$$

$$-8x_3 - 2x_2 + 2x_1 = 0 \quad \text{--- (3)}$$

By Adding eq (1) and (2)

$$2x_3 - 4x_2 - 5x_1 = 0$$

$$-2x_3 - 5x_2 - 4x_1 = 0$$

$$\hline -9x_2 - 9x_1 = 0$$

$$-9(x_2 + x_1) = 0$$

$$x_2 + x_1 = 0$$

let

$$x_1 = 1 \Rightarrow x_2 = -1$$

put these two in eq (3)

$$-8x_3 - 2(-1) + 2(1) = 0$$

$$-8x_3 + 2 + 2 = 0$$

$$-8x_3 + 4 = 0$$

$$8x_3 = 4$$

$$2x_3 = 1$$

$$x_3 = 1/2$$

$$\boxed{x_3 = 0.5}$$

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$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0.5 \end{bmatrix}$$

the unit vector is

$$\underline{e}_1 = \begin{bmatrix} \frac{1}{\sqrt{1^2 + (-1)^2 + (0.5)^2}} \\ \frac{-1}{\sqrt{1^2 + (-1)^2 + (0.5)^2}} \\ \frac{0.5}{\sqrt{1^2 + (-1)^2 + (0.5)^2}} \end{bmatrix} = \begin{bmatrix} 0.667 \\ -0.667 \\ 0.333 \end{bmatrix}$$

$$\underline{e}_1 = \begin{bmatrix} 0.6667 \\ -0.6667 \\ 0.3333 \end{bmatrix}$$

Now,

For $\lambda_2 = 9$

$$\underline{A}\underline{x} = \lambda_2 \underline{x}$$

$$\begin{bmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9x_1 \\ 9x_2 \\ 9x_3 \end{bmatrix}$$

$$13x_1 - 4x_2 + 2x_3 - 9x_1 = 0$$

$$2x_3 - 4x_2 + 4x_1 = 0 \quad \text{--- (c)}$$

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$$-4x_1 + 13x_2 - 2x_3 - 9x_2 = 0$$

$$-2x_3 + 4x_2 - 4x_1 = 0 \quad \text{--- (2)}$$

$$2x_1 - 2x_2 + 10x_3 - 9x_3 = 0$$

$$x_3 - 2x_2 + 2x_1 = 0 \quad \text{--- (3)}$$

Adding eq. (1) and (3)

$$2x_3 + 4x_2 + 4x_1 = 0$$

$$-2x_3 + 4x_2 - 4x_1 = 0$$

Let

$$x_1 = 1, x_2 = 1 \Rightarrow x_3 = 0$$

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

the unit vector is

$$\underline{e}_2 = \begin{bmatrix} \frac{1}{\sqrt{1^2+1^2}} \\ \frac{1}{\sqrt{1^2+1^2}} \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

Similarly,

$$\underline{e}_3 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

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Now, the spectral decomposition is

$$\underline{A} = \lambda_1 \underline{e}_1 \underline{e}_1' + \lambda_2 \underline{e}_2 \underline{e}_2' + \lambda_3 \underline{e}_3 \underline{e}_3'$$

$$\begin{bmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{bmatrix} = 18 \begin{bmatrix} 0.6667 \\ -0.6667 \\ 0.3333 \end{bmatrix} \begin{bmatrix} 0.6667 & -0.6667 & 0.3333 \end{bmatrix}$$

$$+ 9 \begin{bmatrix} 1/3 \\ 1/3 \\ 0 \end{bmatrix} \begin{bmatrix} 1/3 & 1/3 & 0 \end{bmatrix} + 9 \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{bmatrix} = 18 \begin{bmatrix} 0.4445 & -0.4445 & 0.2222 \\ -0.4445 & 0.4445 & -0.2222 \\ 0.2222 & -0.2222 & 0.1111 \end{bmatrix}$$

$$+ 9 \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix} + 9 \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{bmatrix} = \begin{bmatrix} 8.001 & -8.001 & 3.99 \\ -8.001 & 8.001 & -3.99 \\ 3.99 & -3.99 & 1.99 \end{bmatrix} + \begin{bmatrix} 9/2 & 9/2 & 0 \\ 9/2 & 9/2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 9/2 & 9/2 & 0 \\ 9/2 & 9/2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Q. NO. 6

Q. NO. 6

Find a) $E(Y_1/Y_2)$
b) $\text{COV}(Y_1/Y_2)$

Given,

The random vector $X' = [x_1 \ x_2 \ x_3 \ x_4]$

$$U = \begin{bmatrix} 5 \\ 5 \\ 10 \\ 10 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 10 & 4 & 4 & 4 \\ & 20 & 10 & 10 \\ & & 20 & 10 \\ & & & 20 \end{bmatrix}$$

Suppose $Y_1' = [x_1 \ x_2]$ and $Y_2' = [x_3 \ x_4]$

Solution :

$$U = \begin{bmatrix} 5 \\ 5 \\ 10 \\ 10 \end{bmatrix} \Rightarrow \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{matrix}$$

$$\Sigma = \begin{bmatrix} 10 & 4 & 4 & 4 \\ 4 & 20 & 10 & 10 \\ 4 & 10 & 20 & 10 \\ 4 & 10 & 10 & 20 \end{bmatrix}$$

$$\Sigma_{11} = \begin{bmatrix} 10 & 4 \\ 4 & 20 \end{bmatrix}$$

$$\Sigma_{21} = \begin{bmatrix} 4 & 10 \\ 4 & 10 \end{bmatrix}$$

$$\Sigma_{22} = \begin{bmatrix} 4 & 4 \\ 10 & 10 \end{bmatrix}$$

$$\Sigma_{22} = \begin{bmatrix} 20 & 10 \\ 10 & 20 \end{bmatrix}$$

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$$a) E(Y_1/Y_2) = ?$$

$$\tilde{\Sigma}_{22}^{-1} = \frac{\text{Adj } \tilde{\Sigma}_{22}}{|\tilde{\Sigma}_{22}|}$$

$$|\tilde{\Sigma}_{22}| = \begin{vmatrix} 20 & 10 \\ 10 & 20 \end{vmatrix}$$

$$= 400 - 100$$

$$= 300$$

$$\text{Adj } \tilde{\Sigma}_{22} = \begin{bmatrix} 20 & -10 \\ -10 & 20 \end{bmatrix}$$

$$\tilde{\Sigma}_{22}^{-1} = \frac{1}{300} \begin{bmatrix} 20 & -10 \\ -10 & 20 \end{bmatrix}$$

$$= \begin{bmatrix} 0.0666 & -0.0333 \\ -0.0333 & 0.0666 \end{bmatrix}$$

$$E(Y_1/Y_2) = U_1 + \tilde{\Sigma}_{12} \tilde{\Sigma}_{22}^{-1} (Y_2 - U_2)$$

$$= \begin{bmatrix} 5 \\ 5 \end{bmatrix} + \begin{bmatrix} 4 & 4 \\ 10 & 10 \end{bmatrix} \begin{bmatrix} 0.066 & -0.033 \\ -0.033 & 0.066 \end{bmatrix} \begin{bmatrix} Y_2 - 10 \\ Y_2 - 10 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 5 \end{bmatrix} + \begin{bmatrix} 0.12 & 0.12 \\ 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} Y_2 - 10 \\ Y_2 - 10 \end{bmatrix}$$

te:

$$2 \begin{bmatrix} 5 \\ 5 \end{bmatrix} + \begin{bmatrix} 0.24Y_2 - 2.4 \\ 0.6Y_2 - 6 \end{bmatrix}$$

$$= \begin{bmatrix} 0.24Y_2 + 2.6 \\ 0.6Y_2 - 1 \end{bmatrix}$$

$$E(Y_1/Y_2) = \begin{bmatrix} 0.24Y_2 + 2.6 \\ 0.6Y_2 - 1 \end{bmatrix}$$

b) $\text{COV}(Y_1/Y_2)$

$$\text{COV}(Y_1/Y_2) = \Sigma_{12} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

$$= \begin{bmatrix} 10 & 4 \\ 4 & 20 \end{bmatrix} - \begin{bmatrix} 4 & 4 \\ 10 & 10 \end{bmatrix} \begin{bmatrix} 0.066 & -0.033 \\ -0.033 & 0.066 \end{bmatrix} \begin{bmatrix} 4 & 10 \\ 4 & 10 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 4 \\ 4 & 20 \end{bmatrix} - \begin{bmatrix} 0.12 & 0.12 \\ 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 4 & 10 \\ 4 & 10 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 4 \\ 4 & 20 \end{bmatrix} - \begin{bmatrix} 9.04 & 2.4 \\ 2.4 & 6 \end{bmatrix}$$

$$\text{COV}(Y_1/Y_2) = \begin{bmatrix} 9.04 & 1.6 \\ 1.6 & 14 \end{bmatrix}$$

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of LONG QUESTIONS

of Q. NO. 2

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$$

- (1) Find Eigen value and vector.
- (2) Calculate A^{-1} and find eigen value and vector of A^{-1}

Solution

Characteristic equation is

$$|A - \lambda I| = 0$$

$$\left| \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

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$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(-2-\lambda) - 4 = 0$$

$$-2 - \lambda + 2\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 + \lambda - 6 = 0$$

$$\lambda^2 + 3\lambda - 2\lambda - 6 = 0$$

$$\lambda(\lambda + 3) - 2(\lambda + 3) = 0$$

$$(\lambda + 3)(\lambda - 2) = 0$$

$$\lambda = -3$$

$$\lambda = 2$$

$$\lambda_1 = 2$$

$$\lambda_2 = -3$$

$$\text{Trace } A = \lambda_1 + \lambda_2$$

$$1 - 2 = 2 - 3$$

$$-1 = -1 \text{ (OK)}$$

Eigen vector is

$$A \underline{x} = \lambda_1 \underline{x}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 2 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$x_1 + 2x_2 - 2x_1 = 0$$

$$2x_2 - x_1 = 0 \quad \text{--- (1)}$$

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$$2x_1 - 2x_2 - 2x_2 = 0$$

$$-4x_2 + 2x_1 = 0 \quad \text{--- (2)}$$

let

$$x_1 = 2 \Rightarrow x_2 = 1$$

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

The unit vector is

$$\underline{e}_1 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$Ax = x\lambda_2$$

$$\begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 3 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$x_1 + 2x_2 + 3x_1 = 0$$

$$2x_2 + 4x_1 = 0 \quad \text{--- (1)}$$

$$2x_1 - 2x_2 + 3x_2 = 0$$

$$x_2 + 2x_1 = 0 \quad \text{--- (2)}$$

let

$$x_1 = 1, \Rightarrow x_2 = -2$$

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

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The unit vector is .

$$\underline{e}_2 = \begin{bmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix}$$

(ii) Calculate A^{-1} and find Eigen value and vector.

$$A^{-1} = \frac{\text{Adj}(A)}{|A|}$$

$$|A| = \begin{vmatrix} 1 & 2 \\ 2 & -2 \end{vmatrix} = -2 - 4 = -6$$

$$\text{Adj} = \begin{bmatrix} -2 & -2 \\ -2 & 1 \end{bmatrix}$$

$$A^{-1} = -\frac{1}{6} \begin{bmatrix} -2 & -2 \\ -2 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1/3 & 1/3 \\ 1/3 & -1/6 \end{bmatrix} = \underline{\underline{B}}$$

The characteristic equation of $\underline{B}$

$$|\underline{B} - \lambda I| = 0$$

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$$\left| \begin{bmatrix} \frac{1}{3} & \frac{1}{9} \\ \frac{1}{3} & -\frac{1}{6} \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} \frac{1}{3} - \lambda & \frac{1}{9} \\ \frac{1}{3} & -\frac{1}{6} - \lambda \end{vmatrix} = 0$$

$$\left(\frac{1}{3} - \lambda\right)\left(-\frac{1}{6} - \lambda\right) - \frac{1}{9} = 0$$

$$-\frac{1}{18} - \lambda + \lambda + \lambda^2 - \frac{1}{9} = 0$$

$$\lambda^2 - \frac{1}{6} = 0$$

$$-1 - 6\lambda + 3\lambda + 18\lambda - 2 = 0$$

$$18\lambda^2 - 3\lambda - 3 = 0$$

$$6\lambda^2 - \lambda - 1 = 0$$

$$6\lambda^2 - 3\lambda + 2\lambda - 1 = 0$$

$$3\lambda(2\lambda - 1) + 1(2\lambda - 1) = 0$$

$$(2\lambda - 1)(3\lambda + 1) = 0$$

$$2\lambda = 1$$

$$\boxed{\lambda_1 = \frac{1}{2}}$$

$$3\lambda + 1 = 0$$

$$\boxed{\lambda_2 = -\frac{1}{3}}$$

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Eigen vectors.

$$B \underline{x} = \lambda_1 \underline{x}_1$$

$$\begin{bmatrix} 1/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\frac{x_1}{3} + \frac{x_2}{3} - \frac{x_1}{2} = 0$$

multiply '6'

$$2x_1 + 2x_2 - 3x_1 = 0$$

$$2x_2 - x_1 = 0 \quad \text{--- (1)}$$

$$\frac{x_1}{3} - \frac{x_2}{6} - \frac{x_2}{2} = 0$$

multiply '6'

$$2x_1 - x_2 - 3x_2 = 0$$

$$-4x_2 + 2x_1 = 0$$

$$-2x_2 + x_1 = 0 \quad \text{--- (2)}$$

let

$$x_1 = 2, \quad x_2 = 1$$

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\underline{e}_1 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

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$$Bx = \lambda_2 x$$

$$\begin{pmatrix} 1/3 & 1/3 \\ 1/3 & -1/6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -\frac{1}{3} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\frac{x_1}{3} + \frac{x_2}{3} + \frac{x_1}{3} = 0$$

multiply by '3'

$$x_1 + x_2 + x_1 = 0$$

$$x_2 + 2x_1 = 0 \quad \text{--- (1)}$$

$$\frac{x_1}{3} - \frac{x_2}{6} + \frac{x_2}{3} = 0$$

$$2x_1 - x_2 + 2x_2 = 0$$

$$x_2 + 2x_1 = 0$$

let

$$x_1 = 1 \Rightarrow x_2 = -2$$

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

The unit vector is

$$e_2 = \begin{pmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{pmatrix}$$

By Comparing $A = B = A^T$

$\therefore$ $\sqrt{5} e_i$ are orthonormal vectors

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Q. NO. 3

Maximum Likelihood Estimate
of $\underline{\mu}$ & $\underline{\Sigma}$:
of multivariate normal dist.
size 'n'

Proof:

$$L(\underline{\mu}, \underline{\Sigma}) = \prod_{i=1}^n f(\underline{y}_i; \underline{\mu}, \underline{\Sigma})$$

$$L(\underline{\mu}, \underline{\Sigma}) = \prod_{i=1}^n (2\pi)^{-p/2} |\underline{\Sigma}|^{-1/2} \exp\left[-\frac{1}{2}(\underline{y}_i - \underline{\mu})' \underline{\Sigma}^{-1} (\underline{y}_i - \underline{\mu})\right]$$

$$L(\underline{\mu}, \underline{\Sigma}) = (2\pi)^{-np/2} |\underline{\Sigma}|^{-n/2} \exp\left[-\frac{1}{2} \sum_{i=1}^n (\underline{y}_i - \underline{\mu})' \underline{\Sigma}^{-1} (\underline{y}_i - \underline{\mu})\right] \quad (1)$$

Consider,

$$= \sum_{i=1}^n (\underline{y}_i - \underline{\mu})' \underline{\Sigma}^{-1} (\underline{y}_i - \underline{\mu})$$

$$= \sum_{i=1}^n \underbrace{(\underline{y}_i - \underline{\mu})'}_A \underbrace{\underline{\Sigma}^{-1} (\underline{y}_i - \underline{\mu})}_B$$

$$= \sum_{i=1}^n \text{tr} \left[\underline{\Sigma}^{-1} (\underline{y}_i - \underline{\mu}) (\underline{y}_i - \underline{\mu})' \right] \quad \because \text{tr}(AB) = \text{tr}(BA)$$

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$$= \text{de } \sum_{i=1}^n [(y_i - \bar{y})(y_i - \bar{y})'] \quad (2)$$

taking only,

$$= \sum_{i=1}^n (y_i - \bar{y} + \bar{y} - \bar{y})(y_i - \bar{y} + \bar{y} - \bar{y})'$$

$$= \sum_{i=1}^n ((y_i - \bar{y}) + (\bar{y} - \bar{y}))((y_i - \bar{y}) + (\bar{y} - \bar{y})')$$

$$= \sum_{i=1}^n [(y_i - \bar{y}) + (\bar{y} - \bar{y})][(y_i - \bar{y}) + (\bar{y} - \bar{y})']$$

$$= \sum_{i=1}^n [(y_i - \bar{y})(y_i - \bar{y})' + (y_i - \bar{y})(\bar{y} - \bar{y})' + (\bar{y} - \bar{y})(y_i - \bar{y})' + (\bar{y} - \bar{y})(\bar{y} - \bar{y})']$$

$$= \sum_{i=1}^n (y_i - \bar{y})(y_i - \bar{y})' + (\bar{y} - \bar{y})' \sum_{i=1}^n (y_i - \bar{y})$$

$$+ (\bar{y} - \bar{y}) \sum_{i=1}^n (y_i - \bar{y}) + n(\bar{y} - \bar{y})(\bar{y} - \bar{y})'$$

As the sum of deviation from mean is zero $\sum_{i=1}^n (y_i - \bar{y}) = 0$

$$= \sum_{i=1}^n (y_i - \bar{y})(y_i - \bar{y})' + 0 + 0 + n(\bar{y} - \bar{y})(\bar{y} - \bar{y})'$$

Let:

$$= \underline{w} + n(\bar{y} - \underline{u})(\bar{y} - \underline{u})'$$

(B)

Now, substituting eq (2) and (3) in eq (A)

$$L(\underline{\mu}, \underline{\Sigma}) = (2\pi)^{-np/2} |\underline{\Sigma}|^{-n/2} \exp\left\{-\frac{1}{2} \text{tr} \underline{\Sigma}^{-1} \left[\underline{w} + n(\bar{y} - \underline{u})(\bar{y} - \underline{u})' \right]\right\}$$

$$= \frac{1}{(2\pi)^{np/2} |\underline{\Sigma}|^{n/2}} \exp\left\{-\frac{1}{2} \text{tr} \underline{\Sigma}^{-1} \underline{w} - \frac{n}{2} \text{tr} \underline{\Sigma}^{-1} (\bar{y} - \underline{u})(\bar{y} - \underline{u})'\right\}$$

taking a ln on b/s.

$$\ln L(\underline{\mu}, \underline{\Sigma}) = -\frac{np}{2} \ln(2\pi) - \frac{n}{2} \ln |\underline{\Sigma}| - \frac{1}{2} \text{tr} \underline{\Sigma}^{-1} \underline{w} - \frac{n}{2} \text{tr} \underline{\Sigma}^{-1} (\bar{y} - \underline{u})(\bar{y} - \underline{u})'$$

PUACP

It is a scalar quantity

$$= \frac{1}{(2\pi)^{np/2} |\underline{\Sigma}|^{n/2}} \exp\left\{-\frac{n}{2} \left[\text{tr} \underline{\Sigma}^{-1} (\bar{y} - \underline{u})(\bar{y} - \underline{u})' \right]\right\}$$

(4)

Diff w.r.t 'u'

$$\frac{\partial \ln L(\underline{\mu}, \underline{\Sigma})}{\partial \underline{u}} = 0 + 0 + \dots - \frac{n}{2} \left[2 \underline{\Sigma}^{-1} (\bar{y} - \underline{u}) \right]$$

equates it to zero.

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$$-n \left[\Sigma^{-1} (\bar{y} - \mu) \right] = 0$$

$$\Sigma^{-1} (\bar{y} - \mu) = 0$$

$$\bar{y} - \mu = 0$$

$$\boxed{\mu = \bar{y}}$$

It is a maximum likelihood of μ .

Now, Again diff w.r.t Σ^{-1}

$$\frac{\partial \ln(L, \Sigma)}{\partial \Sigma^{-1}} = \frac{n}{2} \left[2(\Sigma^{-1})^{-1} - \text{diag}(\Sigma^{-1})^{-1} \right]$$

$$= \frac{1}{2} \left[\underline{\omega} + \underline{\omega}' - \text{diag}(\underline{\omega}) \right]$$

$$= n \Sigma^{-1} - \frac{n}{2} \text{diag} \Sigma^{-1} - \frac{1}{2} \left[2\underline{\omega} - \text{diag}(\underline{\omega}) \right]$$

$$= n \Sigma^{-1} - \frac{n}{2} \text{diag} \Sigma^{-1} - \underline{\omega} + \frac{1}{2} \text{diag}(\underline{\omega})$$

equating it to zero.

$$\Rightarrow n \Sigma^{-1} - \frac{n}{2} \text{diag} \Sigma^{-1} - \underline{\omega} + \frac{1}{2} \text{diag}(\underline{\omega}) = 0$$

e:

~~Also~~

$$n \left[\underline{\underline{\Sigma}} - \frac{1}{2} \text{diag} \underline{\underline{\Sigma}} \right] = \underline{\underline{w}} - \frac{1}{2} \text{diag} (\underline{\underline{w}})$$

$$\underline{\underline{\Sigma}} - \frac{1}{2} \text{diag} \underline{\underline{\Sigma}} = \frac{1}{n} \left[\underline{\underline{w}} - \frac{1}{2} \text{diag} (\underline{\underline{w}}) \right]$$

The off diag entries is

$$\underline{\underline{\Sigma}}(0) = \frac{1}{n} \left[\underline{\underline{w}} - \frac{1}{2} (0) \right]$$

$$\boxed{\begin{matrix} \underline{\underline{\Sigma}} = \frac{1}{n} \underline{\underline{w}} \\ \underline{\underline{\theta}} = \frac{1}{n} \underline{\underline{w}} \end{matrix}} \quad \text{--- (5)}$$

The Diag entries is

$$\underline{\underline{\Sigma}} - \frac{1}{2} \underline{\underline{\Sigma}} = \frac{1}{n} \left[\underline{\underline{w}} - \frac{1}{2} \underline{\underline{w}} \right]$$

$$\frac{1}{2} \underline{\underline{\Sigma}} = \frac{1}{n} \left[\frac{1}{2} \underline{\underline{w}} \right]$$

$$\boxed{\underline{\underline{\theta}} = \frac{1}{n} \underline{\underline{w}}} \quad \text{--- (6)}$$

So from (5) and (6).

$$\boxed{\underline{\underline{\Sigma}} = \frac{1}{n} \underline{\underline{w}}}$$

Date: _____

Q. NO. 4

$$X = \begin{matrix} & \begin{matrix} X_1 & X_2 & X_3 \end{matrix} \\ \begin{bmatrix} 1 & 2 & 5 \\ 4 & 1 & 5 \\ 4 & 0 & 4 \end{bmatrix} & \begin{matrix} X_1 \\ X_2 \\ X_3 \end{matrix} \end{matrix}$$

- (i) Sample means of $b^T X$ and $C^T X$
(ii) Sample variance and Covariance
of $b^T X$ and $C^T X$.

where,

$$b^T X = [2 \ 2 \ -1] [X_1 \ X_2 \ X_3]$$

$$C^T X = [1 \ -1 \ 3] [X_1 \ X_2 \ X_3]$$

Solution, — PUACP

For Sample Means,

$$\begin{aligned} E[b^T X] &= b^T E(X) \\ &= b^T [E(X_1) + E(X_2) + E(X_3)] \end{aligned}$$

$$\bar{X}_1 = \frac{1+4+4}{3} = 3$$

$$\bar{X}_2 = \frac{2+1+0}{3} = 1$$

$$\bar{X}_3 = \frac{5+5+4}{3} = 4.667$$

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$$F(b^T X) = b^T [3 + 1 + 4.667]$$

$$\text{Means} = [2 \quad 2 \quad -1] [8.667]$$

$$\text{Means} = [17.334 \quad 17.334 \quad -8.667]$$

$$b^T F(X_1) = 17.334$$

$$b^T F(X_2) = 17.334$$

$$b^T F(X_3) = -8.667$$

For $C^T X$

$$C^T X = [1 \quad -1 \quad 3] [X_1 \quad X_2 \quad X_3]$$

$$= [1 \quad -1 \quad 3] [-8.667]$$

$$C^T X = [-8.667 \quad 8.667 \quad 26.001]$$

$$C^T F(X_1) = -8.667$$

$$C^T F(X_2) = 8.667$$

$$C^T F(X_3) = 26.001$$

Date: _____

ii) Sample variance and covariance.

Formula:-

$$S = (n-1)^{-1} \sum_{i=1}^n (\underline{x}_i \underline{x}_i' - n \underline{\bar{x}} \underline{\bar{x}}')$$

$$\therefore n = 3,$$

$$= (3-1)^{-1} \left[\begin{bmatrix} 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix} + \begin{bmatrix} 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 5 \end{bmatrix} + \begin{bmatrix} 4 & 0 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 4 \end{bmatrix} - 3 \begin{bmatrix} 3 & 1 & 4.667 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 4.667 \end{bmatrix} \right]$$

$$= 2^{-1} \left[\begin{bmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \\ 5 & 10 & 25 \end{bmatrix} + \begin{bmatrix} 16 & 4 & 20 \\ 4 & 1 & 5 \\ 20 & 5 & 25 \end{bmatrix} + \begin{bmatrix} 16 & 0 & 16 \\ 0 & 0 & 0 \\ 16 & 0 & 16 \end{bmatrix} - 3 \begin{bmatrix} 9 & 3 & 14.001 \\ 3 & 1 & 4.667 \\ 14.001 & 4.667 & 21.78 \end{bmatrix} \right]$$

$$= \frac{1}{2} \begin{bmatrix} 1+16+16 & 2+4+0 & 5+20+16 \\ 2+4+0 & 4+1+0 & 10+5+0 \\ 5+20+16 & 10+5+0 & 25+25+16 \end{bmatrix} - \begin{bmatrix} 27 & 9 & 42.003 \\ 9 & 3 & 14.001 \\ 42.003 & 14.001 & 68.34 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 33 & 6 & 41 \\ 6 & 5 & 15 \\ 41 & 15 & 66 \end{bmatrix} - \begin{bmatrix} 27 & 9 & 42.003 \\ 9 & 3 & 14.001 \\ 42.003 & 14.001 & 68.34 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 6 & -3 & -1.003 \\ -3 & 2 & 0.999 \\ -1.003 & 0.999 & 0.66 \end{bmatrix} = \begin{bmatrix} 3 & -1.5 & -0.5 \\ -1.5 & 1 & 0.5 \\ -0.5 & 0.5 & 0.33 \end{bmatrix}$$

$$\text{Var}(X_1) = 3$$

$$\text{Var}(X_2) = 1$$

$$\text{Var}(X_3) = 0.33$$

not complete

$$\text{Cov}(X_1, X_2), \text{Cov}(X_2, X_1) = -1.5$$

$$\text{Cov}(X_1, X_3), \text{Cov}(X_3, X_1) = -0.5$$

$$\text{Cov}(X_2, X_3), \text{Cov}(X_3, X_2) = 0.5$$

Date: _____

Q. No. 5

Any two properties of Wishart distⁿ.

- ⇒ Linear Combination
- ⇒ Linear Compound



~~2021~~

~~(Q.NO.2)~~

— (a) —

Convert the covariance matrix

$$\Sigma = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 9 & -3 \\ 2 & -3 & 25 \end{bmatrix}$$

to a correlation matrix.

Solution :-

The sample correlation matrix is

$$\underline{R} = \underline{D}^{-1/2} \underline{\Sigma} \underline{D}^{-1/2}$$

$$\underline{D} = \text{Diag}(\underline{\Sigma})$$

$$\underline{D} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 25 \end{bmatrix}$$

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$$D^{-1/2} = \begin{bmatrix} (4)^{-1/2} & 0 & 0 \\ 0 & (9)^{-1/2} & 0 \\ 0 & 0 & (25)^{-1/2} \end{bmatrix}$$

$$D^{1/2} = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 8.5 & 0 \\ 0 & 0 & 0.2 \end{bmatrix}$$

$$\tilde{R} = D^{-1/2} S D^{1/2}$$

$$\tilde{R} = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 8.5 & 0 \\ 0 & 0 & 0.2 \end{bmatrix} \begin{bmatrix} 4 & 1 & 2 \\ 1 & 9 & -3 \\ 2 & -3 & 25 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 8.5 & 0 \\ 0 & 0 & 0.2 \end{bmatrix}$$

$$\tilde{R} = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 8.5 & 0 \\ 0 & 0 & 0.2 \end{bmatrix} \begin{bmatrix} 2 & 8.5 & 0.4 \\ 0.5 & 76.5 & -0.6 \\ 1 & -25.5 & 5 \end{bmatrix}$$

$$\tilde{R} = \begin{bmatrix} 1 & 4.25 & 0.2 \\ 4.25 & 65.025 & -5.1 \\ 0.2 & -5.1 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$\tilde{R} = \begin{bmatrix} 1 & 4.25 & 0.2 \\ 4.25 & 65.025 & -5.1 \\ 0.2 & -5.1 & 1 \end{bmatrix}$$

Date: _____

— (b) —

Given,

$$\lambda_1 = 10 \quad \lambda_2 = 5 \quad , \quad \lambda_3 = 2$$

$$e_1 = \begin{bmatrix} 1/\sqrt{6} & 2/\sqrt{6} & 1/\sqrt{6} \end{bmatrix}^T$$

$$e_2 = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} & 0 \end{bmatrix}^T$$

$$e_3 = \begin{bmatrix} 1/\sqrt{30} & 2/\sqrt{30} & -5/\sqrt{30} \end{bmatrix}^T$$

what would be the eigen value and
eigen vector of $B = A^3$

Solution:

As we know that the
square root matrix is

$$\underline{A} = \underline{P} \underline{\Lambda} \underline{P}'$$

where,

$$B = A^3$$

So,

$$\underline{A}^3 = \underline{P} \underline{\Lambda}^3 \underline{P}' \quad \text{--- (A)}$$

Now,

$$\underline{\Lambda}^3 = \begin{bmatrix} (10)^3 & 0 & 0 \\ 0 & (5)^3 & 0 \\ 0 & 0 & (2)^3 \end{bmatrix}$$

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$$\underline{A}^3 = \begin{bmatrix} 1000 & 0 & 0 \\ 0 & 125 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

And,

$$\underline{P} = [\underline{e}_1 \quad \underline{e}_2 \quad \underline{e}_3]$$

$$\underline{P} = \begin{bmatrix} 1/\sqrt{6} & 2/\sqrt{5} & 1/\sqrt{30} \\ 2/\sqrt{6} & -1/\sqrt{5} & 2/\sqrt{30} \\ 1/\sqrt{6} & 0 & -5/\sqrt{30} \end{bmatrix}$$

putting the values in eq (A)

$$\underline{A}^3 = \underline{P} \underline{A}^3 \underline{P}^T$$

$$\underline{A}^3 = \begin{bmatrix} 1/\sqrt{6} & 2/\sqrt{5} & 1/\sqrt{30} \\ 2/\sqrt{6} & -1/\sqrt{5} & 2/\sqrt{30} \\ 1/\sqrt{6} & 0 & -5/\sqrt{30} \end{bmatrix} \begin{bmatrix} 1000 & 0 & 0 \\ 0 & 125 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1/\sqrt{6} & 2/\sqrt{6} & 1/\sqrt{6} \\ 2/\sqrt{5} & -1/\sqrt{5} & 0 \\ 1/\sqrt{30} & 2/\sqrt{30} & -5/\sqrt{30} \end{bmatrix}$$

$$\underline{A}^3 = \begin{bmatrix} 1000/\sqrt{6} & 250/\sqrt{5} & 8/\sqrt{30} \\ 2000/\sqrt{6} & -125/\sqrt{5} & 16/\sqrt{30} \\ 1000/\sqrt{6} & 0 & -40/\sqrt{30} \end{bmatrix} \begin{bmatrix} 1/\sqrt{6} & 2/\sqrt{6} & 1/\sqrt{6} \\ 2/\sqrt{5} & -1/\sqrt{5} & 0 \\ 1/\sqrt{30} & 2/\sqrt{30} & -5/\sqrt{30} \end{bmatrix}$$

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$$\tilde{A}^3 = \begin{bmatrix} 267 & 284 & 165.33 \\ 284 & 693 & 331 \\ 165.33 & 331 & 173.33 \end{bmatrix}$$

Answer

~~Ex(C)32~~

for a data matrix.

$$X = \begin{bmatrix} 7 & 9 \\ 3 & 4 \\ 5 & 5 \end{bmatrix}$$

COV = ?

Soln

$$\bar{x}_1 = \frac{7+3+5}{3} = \frac{15}{3} = 5$$

$$\bar{x}_2 = \frac{9+4+5}{3} = \frac{18}{3} = 6$$

Formula,

$$\tilde{X} = (n-1)^{-1} \sum_{i=1}^n (x_i x_i' - n \bar{x} \bar{x}')$$

Date: _____

$$= (3-1) \left[\begin{bmatrix} 7 \\ 9 \end{bmatrix} \begin{bmatrix} 7 & 9 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 \\ 5 \end{bmatrix} \begin{bmatrix} 5 & 5 \end{bmatrix} - 3 \begin{bmatrix} 5 \\ 6 \end{bmatrix} \begin{bmatrix} 5 & 6 \end{bmatrix} \right]$$

$$= (2)^{-1} \left[\begin{bmatrix} 49 & 63 \\ 63 & 81 \end{bmatrix} + \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix} + \begin{bmatrix} 25 & 25 \\ 25 & 25 \end{bmatrix} - 3 \begin{bmatrix} 25 & 30 \\ 30 & 36 \end{bmatrix} \right]$$

$$= \frac{1}{2} \left[\begin{bmatrix} 83 & 100 \\ 100 & 129 \end{bmatrix} - \begin{bmatrix} 75 & 90 \\ 90 & 108 \end{bmatrix} \right]$$

$$= \frac{1}{2} \begin{bmatrix} 8 & 10 \\ 10 & 21 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 5 & 7 \end{bmatrix}$$

$$\text{COV}(X) = 5$$

e:

— (d) —

$$X = [x_1, x_2, x_3, x_4]^T$$

$$U = [4 \ 3 \ 2 \ 1]^T$$

if partitioned as

$$X = [x_1 \ x_2 \mid x_3 \ x_4]^T$$

$$x = [x_1 \mid x_2]^T$$

$$a = [1 \ 2]^T$$

$$E(ax_1) = ?$$

this is wrong procedure

the solution in definition PDF

next pg

Solution :-

$$E(ax_1) = a E(x_1)$$

$$= a [E(x_1) + E(x_2)]$$

$$U = [4 \ 3 \mid 2 \ 1]^T$$

$$U = \begin{pmatrix} 4 \\ 3 \\ 2 \\ 1 \end{pmatrix} \begin{matrix} \rightarrow x_1 \\ x_2 \\ \\ \end{matrix}$$

Date: _____

$$\begin{aligned}\text{So, } E(ax) &= \begin{bmatrix} 1 & 2 \end{bmatrix}^T [3+4] \\ &= \begin{bmatrix} 1 \\ 2 \end{bmatrix} (7) \\ &= \begin{bmatrix} 7 \\ 14 \end{bmatrix}\end{aligned}$$

Question (3)

Same in (2017).

Question (5)

Same in 2017, 2018

— (d) —

$$X = [x_1 \ x_2 \ x_3 \ x_4]^T$$

$$U = [4 \ 3 \ 2 \ 1]^T$$

if partitioned as

$$X = [x_1 \ x_2 | x_3 \ x_4]^T$$

$$x = [x_1 | x_2]^T$$

$$a = [1 \ 1]$$

$$E(ax_1) = ?$$

Solution :-

PUACP

(Wrong)

$$\begin{aligned} E(ax_1) &= a E(x_1) \\ &= a [E(x_1) + E(x_2)] \end{aligned}$$

$$\Rightarrow U = [4 \ 3 | 2 \ 1]^T$$

$$U = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} \begin{matrix} x_1 \\ x_2 \\ \\ \end{matrix}$$

Date: _____

(2021)

— (d) —

$$E(aX_1) = ?$$

Sol:-

$$E(aX_1) = a E(X_1)$$

$$= a E \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

1) $X_1 = [X_1 \ X_2]$ *وہاں پر X_1 اور X_2 کے لیے*

$X_2 = [X_1 \ X_2]$ *وہاں پر X_1 اور X_2 کے لیے*

divide f_1 *وہاں پر f_1 کے لیے*

divide f_2 *وہاں پر f_2 کے لیے*

PUACP

$$U = \begin{bmatrix} X_1 & X_2 & X_3 \\ 4 & 3 & 2 & 1 \end{bmatrix}$$

X_1

Hence

$$= a \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\text{Mean} = 4 + 6 = 7$$

Date: _____

(2021)

— (d) —

$$E(ax_1) = ?$$

Sol:

$$\begin{aligned} E(ax_1) &= a E(x_1) \\ &= a E \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

1) $x_1 = [x_1 \ x_2]$ (row vector)

$x_2 = [x_3 \ x_4]$ (row vector)

divide by 4, 3, 2, 1 respectively
divide by 4, 3, 2, 1 respectively
PUACP

$$U = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ 4 & 3 & 2 & 1 \end{bmatrix}$$

Hence

$$= a \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\text{Mean} = 4 + 6 = 10$$

MULTIVARIATE

Generalized Variance:-

The generalized variance is a scalar value which generalize the variance for multivariate random variable. It was introduced by "Samuel Wilks."

The generalized variance is define as the determinant of covariance. $\det|\Sigma|$

$$\text{Generalized variance} = |\Sigma| = \lambda_1 \cdot \lambda_2 \cdot \lambda_3 \dots \lambda_n$$

date: _____

Decomposition of Matrix

A matrix decomposition is a way of reducing a matrix into its constituent (different) parts.

Let A be a $k \times k$ symmetric matrix. Then A can be expressed in terms of its k eigen values and eigen vectors pairs (λ_i, P_i) .

$$A = \lambda_1 \underline{e}_1 \underline{e}_1' + \lambda_2 \underline{e}_2 \underline{e}_2' + \dots + \lambda_k \underline{e}_k \underline{e}_k'$$

Date: _____

Decomposition of Matrix

A matrix decomposition is a way of reducing a matrix into its constituent (different) parts.

Let 'A' a $K \times K$ symmetric matrix then 'A' can be expressed in term of its K eigen values and eigen vectors pairs (λ_i, e_i) .

$$A = \lambda_1 e_1 e_1' + \lambda_2 e_2 e_2' + \dots + \lambda_K e_K e_K'$$

Date: _____

Wishart distribution:-

Wishart distⁿ is a multivariate version of χ^2 -distribution OR more precisely of the $\delta^2 \chi^2$ -dist.

Since

$$\sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{y_i - \mu}{\sigma} \right)^2 \sim \chi^2(n)$$

where $Z_i \sim N(0, 1)$

$$\sum_{i=1}^n (y_i - \mu)^2 \sim \chi^2(n)$$

Also

$$\sum_{i=1}^n (y_i - \bar{y})^2 \sim \delta^2 \chi^2(n-1)$$

where $y_i \sim N(\mu, \sigma^2)$

Multivariate Normal Dist.

A multivariate normal distⁿ is a vector in multiple normally distributed

ates:

variables. Such that any linear combination of variable

Proof:-

The p.d.f of ^{mult.} normal dist is

$$f(x) = \frac{1}{| \Sigma |^{1/2} (2\pi)^{p/2}} e^{-\frac{1}{2} (x-\mu)' \Sigma^{-1} (x-\mu)}$$

$$-\infty < x_i < \infty$$

Positive Definite

The symmetric matrix A is positive definite if for any vector $x \neq 0$

The quadratic form.

$$x' A x > 0$$

Semi Definite:-

A symmetric matrix A is semi definite if

$$x' A x \geq 0$$

ate: _____

variables. Such that any linear combination of variable

proof:-

The p.d.f of ^{mult.} normal dist is

$$f(x) = \frac{1}{|\Sigma|^{1/2} (2\pi)^{p/2}} e^{-\frac{1}{2} \{ (x-\mu)' \Sigma^{-1} (x-\mu) \}}$$

$$-\infty < x_i < \infty$$

Positive Definite

The symmetric matrix A is said to be positive definite if for any vector $x \neq 0$ the quadratic form $x'Ax > 0$.

Semi Definite:-

A symmetric matrix A is said to be semi-definite if $x'Ax \geq 0$.

Date: _____

Data Matrix.

In multivariate analysis X is called data matrix. Data matrix could be written as the vector of vectors.

i.e

$$\begin{bmatrix} X_{11} & X_{12} & \dots & X_{1p} \\ X_{21} & X_{22} & \dots & X_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{np} & \dots & X_{np} \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{bmatrix}$$

X or X (bold) used for matrix.

PUACP

Eigen Values.

Let 'A' be a $k \times k$ matrix and 'I' is a $k \times k$ identity matrix then the scalar $x_1, x_2, x_3, \dots, x_k$

ate: _____

satisfying the polynomial equation

$$|A - \lambda I| = 0$$

are called eigen values of matrix A .

Eigen Vector

Let A be a $k \times k$ square matrix and λ be an eigen value of matrix A . If x is a non-zero vector such that

$$Ax = \lambda x$$

then x is called the eigen vector of A .